DECODELABS

Project 3: E-Commerce Sales Analysis Using SQL

Data Analytics Internship | SkillAhead Solutions Ltd

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Executive Summary

This report presents the findings of an SQL-based exploratory analysis conducted on an e-commerce sales dataset as part of the DecodeLabs Data Analytics Internship. The analysis covers 1,200 transactions across 14 fields, using 18 structured queries to extract insights across revenue performance, customer behavior, product performance, marketing effectiveness, and order fulfillment. The objective is to support data-driven business decision-making through structured SQL analysis.

1. Dataset Overview

Property	Detail
Dataset Name	Orders Table
Total Records	1,200 Orders
Database Management System	MySQL
Database Name	Project_3
Total Columns	14

1.1 Key Fields

Column	Description
OrderID	Unique identifier for each order
Date	Date the order was placed
CustomerID	Unique identifier for each customer

Product	Product purchased
Quantity	Number of units ordered
UnitPrice	Price per unit
ShippingAddress	Delivery address
PaymentMethod	Method of payment used
OrderStatus	Current status of the order
TrackingNumber	Shipment tracking number
ItemsInCart	Total items in the customer cart
CouponCode	Promotional coupon applied
ReferralSource	Marketing channel that referred the customer
TotalPrice	Total transaction value

2. SQL Environment

The analysis was conducted using MySQL. The database was set up as follows:

```
CREATE DATABASE Project_3;
USE Project_3;
```

3. Data Inspection & Cleaning

3.1 Initial Data Inspection

Query	SQL	Purpose
View Dataset	SELECT * FROM orders_table;	Verify successful data load
Count Total Orders	SELECT COUNT(*) AS Total_Orders FROM orders_table;	Confirm 1,200 records
Preview First 10 Rows	SELECT * FROM orders_table LIMIT 10;	Check data structure

3.2 Unique Values Check

Distinct values were inspected across all categorical columns to identify quality issues:

```
SELECT DISTINCT Product FROM orders_table;
SELECT DISTINCT PaymentMethod FROM orders_table;
SELECT DISTINCT OrderStatus FROM orders_table;
```

```
SELECT DISTINCT CouponCode FROM orders_table;
SELECT DISTINCT ReferralSource FROM orders_table;
```

Finding: Missing values were discovered in the CouponCode field.

3.3 Missing Values — CouponCode Column

309 records had empty CouponCode values. Investigation confirmed these were not errors — they indicated no coupon was used during the transaction.

Resolution:

```
UPDATE orders_table
SET CouponCode = 'No Coupon Used'
WHERE CouponCode = '';
```

Change ID	Description	Impact	Status
CR001	Replaced 309 empty CouponCode values with 'No Coupon Used'	309 records preserved	Resolved

4. Revenue Analysis

4.1 Total Revenue Generated

```
SELECT SUM(ROUND(TotalPrice,0)) AS Total_Revenue
FROM orders_table;
```

Metric	Result
Total Revenue	$1,264,738

Business Insight: The business generated over $1.26 million in total sales revenue.

Recommendation: Continue investing in high-performing products and top acquisition channels.

4.2 Average Order Value

```
SELECT AVG(ROUND(TotalPrice,-1)) AS Avg_Order_Value
FROM orders_table;
```

Metric	Result
Average Order Value	$1,053.95

Business Insight: Customers spend approximately $1,054 per transaction on average.

Recommendation: Introduce product bundles and upselling strategies to push this figure higher.

5. Customer Analysis

```
SELECT COUNT(DISTINCT CustomerID) AS Total_Customers
FROM orders_table;
```

Metric	Result
Total Unique Customers	1,189

Business Insight: The business attracted 1,189 unique customers across 1,200 orders.

Recommendation: Develop customer retention strategies to encourage repeat purchases. With 1,200 orders but only 1,189 customers, the repeat purchase rate is currently very low.

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6. Product Analysis

6.1 Revenue by Product

```
SELECT Product,
SUM(ROUND(TotalPrice)) AS Total_Revenue
FROM orders_table
GROUP BY Product
ORDER BY Total_Revenue DESC;
```

Business Insight: Products generating the highest revenue represent the company's strongest performers.

Recommendation: Increase inventory and marketing efforts for top-performing products.

6.2 Quantity Sold by Product

```
SELECT Product,
SUM(Quantity) AS Total_Quantity
FROM orders_table
GROUP BY Product
ORDER BY Total_Quantity DESC;
```

Business Insight: Highest-selling products reveal customer demand patterns.

Recommendation: Maintain adequate stock levels for high-volume products to avoid missed sales.

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7. Payment Method Analysis

```
SELECT PaymentMethod,
SUM(ROUND(TotalPrice)) AS Total_Revenue
```

```
FROM orders_table
GROUP BY PaymentMethod
ORDER BY Total_Revenue DESC;
```

Business Insight: All 5 payment methods show near-equal usage — no single method dominates.

Recommendation: Ensure a seamless user experience across all payment options to avoid friction at checkout.

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8. Marketing & Referral Analysis

8.1 Revenue by Referral Source

```
SELECT ReferralSource,
SUM(ROUND(TotalPrice)) AS Total_Revenue
FROM orders_table
GROUP BY ReferralSource
ORDER BY Total_Revenue;
```

Business Insight: Instagram is the top revenue-generating channel. Marketing channels differ significantly.

Recommendation: Allocate larger marketing budgets to Instagram and highest-performing referral channels.

8.2 Average Cart Size by Referral Source

```
SELECT ReferralSource,
AVG(ROUND(ItemsInCart,0)) AS Avg_Cart_Size
FROM orders_table
GROUP BY ReferralSource
ORDER BY Avg_Cart_Size DESC;
```

Business Insight: Certain channels attract customers who purchase more items per order.

Recommendation: Focus acquisition on channels that produce larger cart sizes, not just higher order counts.

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9. Order Fulfillment Analysis

9.1 Orders by Status

```
SELECT OrderStatus,
COUNT(*) AS Total_Orders
FROM orders_table
GROUP BY OrderStatus
```

```
ORDER BY Total_Orders;
```

Status	Count	Percentage
Cancelled	250	20.8%
Delivered	231	19.3%
Pending	237	19.8%
Returned	247	20.6%
Shipped	235	19.6%

9.2 Revenue Lost to Returns

```
SELECT SUM(ROUND(TotalPrice)) AS Returned_Value
FROM orders_table
WHERE OrderStatus = 'Returned';
```

Metric	Value
Revenue Lost to Returns	$243,268

Business Insight: Returns account for a significant portion of lost revenue.

Recommendation: Analyze product quality, shipping performance, and customer expectations to reduce return rates.

9.3 Revenue Lost to Cancellations

```
SELECT SUM(ROUND(TotalPrice)) AS Cancelled_Value
FROM orders_table
WHERE OrderStatus = 'Cancelled';
```

Metric	Value
Revenue Lost to Cancellations	$276,390
Revenue Lost to Returns	$243,268
Total Revenue at Risk	$519,658

Business Insight: Over $519,000 in revenue was lost to cancellations and returns combined — 41% of total revenue.

Recommendation: Review checkout experience, payment processing, and order fulfillment speed.

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10. Promotional Analysis

```
SELECT CouponCode,
COUNT(*) AS Usage_Count
FROM orders_table
GROUP BY CouponCode
ORDER BY Usage_Count;
```

Business Insight: Popular coupon codes indicate which promotions resonate most with customers.

Recommendation: Replicate successful promotional campaigns and retire underperforming offers.

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11. Revenue Trend Analysis

```
SELECT YEAR(Date) AS Order_Year,
SUM(ROUND(TotalPrice,0)) AS Revenue
FROM orders_table
GROUP BY YEAR(Date)
ORDER BY Order_Year;
```

Business Insight: Yearly revenue trends reveal business growth and decline patterns over time.

Recommendation: Use historical trends to support future sales forecasting, budgeting, and inventory planning.

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12. Key Findings Summary

Finding	Value
Total Revenue	$1,264,738
Average Order Value	$1,053.95
Unique Customers	1,189
Revenue Lost to Returns	$243,268
Revenue Lost to Cancellations	$276,390
Combined Revenue at Risk	$519,658
Cancellation Rate	20.8%
Return Rate	20.6%
Combined Loss Rate	41.4%

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13. Final Recommendations

1. Increase focus on top-performing products — prioritize inventory and marketing.
2. Optimize marketing spend toward high-revenue referral sources.
3. Reduce cancellations through improved order processing and checkout experience.
4. Investigate return causes — product quality, shipping, and customer expectations.
5. Expand successful promotional campaigns based on coupon usage data.
6. Develop customer retention strategies to increase repeat purchase rate.
7. Use revenue trends for forecasting and inventory planning.

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14. Conclusion

Through structured SQL queries, raw transactional data was transformed into meaningful business insights. The analysis identified over $1.26 million in total revenue, highlighted a critical 41.4% cancellation and return rate, revealed Instagram as the top-performing marketing channel, and uncovered a consistent downward revenue trend that warrants immediate strategic attention.

The findings from this analysis provide a clear foundation for data-driven decision-making across product strategy, marketing investment, operational improvement, and customer retention.

Sign-Off

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Raw data. Real queries. Actionable insights.